

④ a) Location: 2 bytes = 16 bits  $\rightarrow 2^{16} - 1$  free.

$$A = (9A7D)_{16} = 1001\ 1010\ 0111\ 1101$$

$$B = (7583)_{16} = 0111\ 0101\ 1000\ 0011$$

$A = (9A7D)_{16} \rightarrow 1$ 's complement:

$$0110\ 0101\ 1000\ 0010$$

$\rightarrow 2$ 's complement:

$$0110\ 0101\ 1000\ 0010 +$$

1

$$\hline 0110\ 0101\ 1000\ 0011$$

$B = (7583)_{16} \rightarrow 1$ 's:

$$1000\ 1010\ 0111\ 1100$$

$\rightarrow 2$ 's:

$$1000\ 1010\ 0111\ 1100 +$$

1

$$\hline 1000\ 1010\ 0111\ 1101$$

- $A \& B$  are 1's complements of each other only if  $B = 1$ 's complement of  $A$ . X
- a — 11 — 2's complements of each other if  $B = 2$ 's complement of  $A$ . X

b) Location: 4 bytes = 32 bits  
 $A = (00F095D)_{16} = 000\ 1111\ 0\ 1001\ 0101\ 1101$   
 $B = (FFF0F6A3)_{16} = 1111\ 1111\ 1111\ 0\ 1111\ 0110\ 1010\ 0010$

$A \rightarrow 1$ 's: 111

$$A = 0000\ 0000\ 0000\ 1111\ 0000\ 1001\ 0101\ 1101$$

$$B = 1111\ 1111\ 1111\ 0000\ 1111\ 0110\ 1010\ 0010$$

$$A \rightarrow 1's : 1111\ 1111\ 1111\ 0000\ 1111\ 0110\ 1010\ 0010$$

$$2's : 1111\ 1111\ 1111\ 0000\ 1111\ 0110\ 1010\ 0011$$

$$B \rightarrow 1's : 0000\ 0000\ 0000\ 1111\ 0000\ 1001\ 0101\ 1101$$

$$2's : 0000\ 0000\ 0000\ 1111\ 0000\ 1001\ 0101\ 1110$$

$B = 1$ 's complement of  $A$

$\Rightarrow A \& B$  are 1's complements ✓

$B \neq 2$ 's complement of  $A$

?  $\Rightarrow A \& B$  are not 2's complements



c) Location: 2 bytes = 16 bits.

$$A = (4BA1)_{16}$$

$$= 0100\ 1011\ 1010\ 0001\ (2)$$

$$A \rightarrow \underline{1's}: 1011\ 0100\ 0101\ 1110$$

$$\underline{2's}: 1011\ 0100\ 0101\ 1111$$

$$\cancel{B} \rightarrow 1's\ B = (5C93)_{16}$$

$$= 0101\ 1100\ 1001\ 0011\ (2)$$

$$B \rightarrow \underline{1's}: 1011\ 0011\ 0110\ 1100$$

$$\underline{2's}: 1011\ 0011\ 0110\ 1101$$

$$B \neq 1's\ X$$

$$B \neq 2's\ X$$

A number can be both 1's complement and 2's complement w another number?

$$\begin{array}{cccccccc} 0001, & 0010, & 0011, & 0100, & 0101, & 0110, & 0111, & \\ \overset{2^0}{1}, & \overset{2^1}{0}, & & \overset{2^2}{0}, & \overset{2^3}{1}, & \overset{2^4}{0}, & \overset{2^5}{1}, & \overset{2^6}{1}, \\ \overset{2^7}{1}, & 1001, & 1010, & 1011, & 1100, & 1101, & 1110, & \\ \overset{8}{1111}, & & & \overset{11}{11}, & & & & \end{array}$$

d) Location: 1 byte: 8 bits.

$$A = (7F)_{16} = 0111\ 1111\ (2)$$

$$A \rightarrow \underline{1's}: 1000\ 0000$$

$$\underline{2's}: 1000\ 0001$$

$$B = (81)_{16} = 1000\ 0001\ (2)$$

$$B \rightarrow \underline{1's}: 0111\ 1110$$

$$\underline{2's}: 0111\ 1111$$

$$B = 2's A$$

$\Rightarrow A \& B$  are 2's complements

e) Location: 2 bytes: ~~16~~ bits

$$A = (F32A)_{16} = 0111\ 0011\ 0010\ 1010\ (2)$$

$$\rightarrow \underline{1's}: 1000\ 1100\ 1101\ 0101\ (2)$$

$$\rightarrow \underline{2's}: 1000\ 1100\ 1101\ 0110\ (2)$$



$$B = (4E58)_{16}$$

$$= 0100\ 1110\ 0101\ 1000(2)$$

$$\rightarrow 1's : 1011\ 0001\ 1010\ 0111$$

$$\rightarrow \underline{\underline{2's}} : 1011\ 0001\ 1010\ 1000$$

$$B \neq 1's$$

$$\neq 2's$$

X



⑧

Unsigned representation on 8 bits:

- 8 → binary: 1000  
→ on 8 bits: 0000 1000

•  $64 \div 2 = 32 \div 2 = 16 \div 2 = 8 \div 2 = 4 \div 2 = 2 \div 2 = 1$   
 $\Rightarrow 01000011$

•  $230 \div 2 = 115 \div 2 = 57 \div 2 = 28 \div 2 = 14 \div 2 = 7 \div 2 = 3 \div 2 = 1$   
 $\Rightarrow 011100110$   
 $\Rightarrow 000011100110$

⑨

on 16 bits:

- -6 → 2's complement

$|-6| = 6 \rightarrow 0000\ 0000\ 0000\ 0110$

$\{ 1's$   
 $1111\ 1111\ 1111\ 1001$

$\{ 2's (+1)$   
 $\downarrow$

$-6(10) = 1111\ 1111\ 1111\ 1010(2)$

- -121 → 2's C:

$|-121| = 121 \rightarrow 121 \div 2$

$12 \div 2 = 6 \div 2 = 3 \div 2 = 1 \div 2 = 0$   
 $60 \div 2 = 30 \div 2 = 15 \div 2 = 7 \div 2 = 3 \div 2 = 1 \div 2 = 0$   
 $30 \div 2 = 15 \div 2 = 7 \div 2 = 3 \div 2 = 1 \div 2 = 0$   
 $15 \div 2 = 7 \div 2 = 3 \div 2 = 1 \div 2 = 0$   
 $7 \div 2 = 3 \div 2 = 1 \div 2 = 0$   
 $3 \div 2 = 1 \div 2 = 0$   
 $1 \div 2 = 0$

$\rightarrow 01111001 \rightarrow 0000\ 0000\ 0111\ 1001$

$\{ 1's$

$\{ 1's$

$1000\ 0110$

$1111\ 1111\ 1000\ 0110$

$\{ 2's (+1)$

$\{ 2's (+1)$

$1000\ 0111$

$1111111\ 1000\ 0111$